

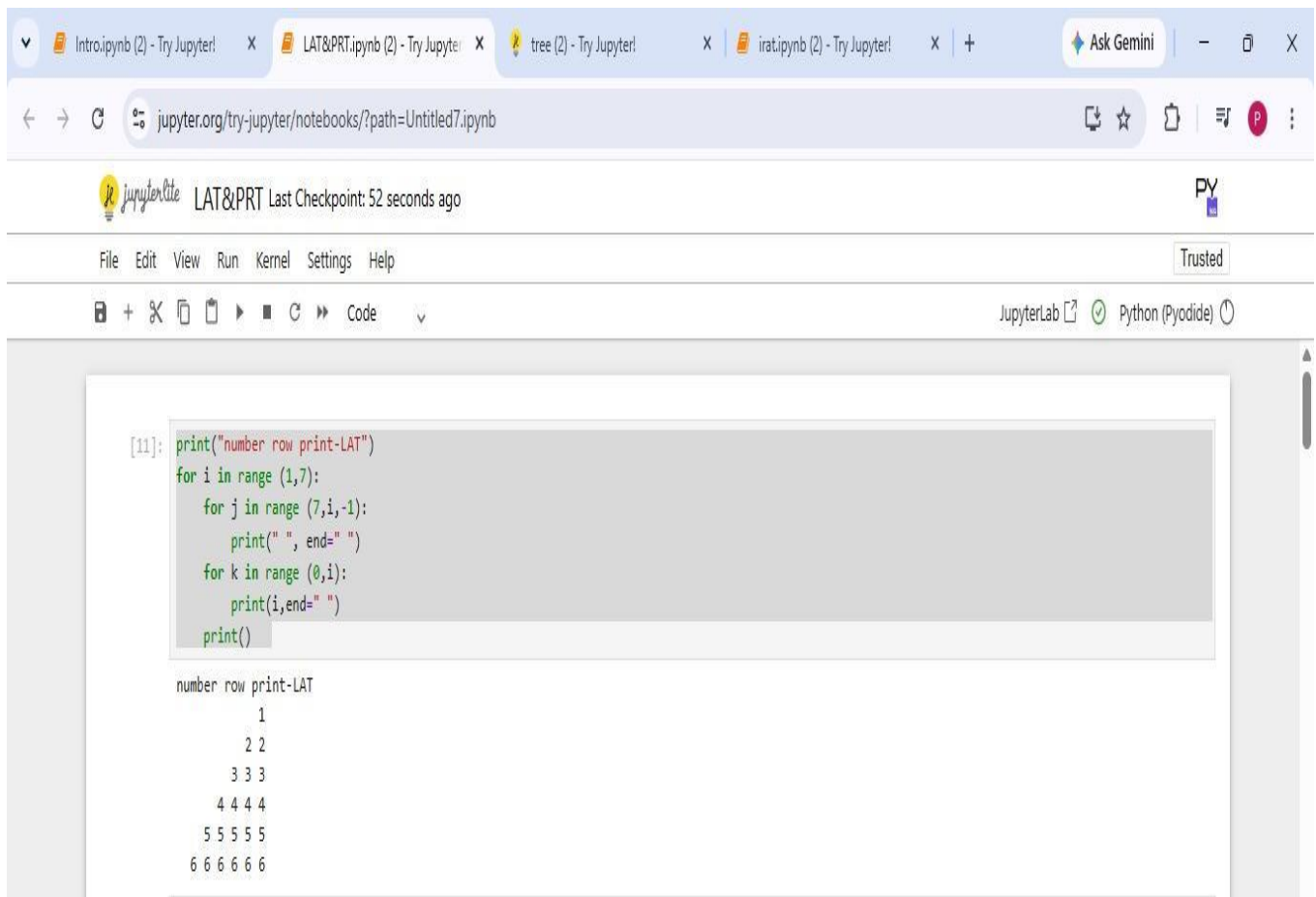
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GEN AI – ASSIGNMENT 4 – VL (03.06.2026)

Topic: LAT&PRT

1.Left Angle Triangle (LAT) – Number- Row

Output



The screenshot shows a JupyterLab interface with a browser window at the top displaying the URL `jupyter.org/try-jupyter/notebooks/?path=Untitled7.ipynb`. The JupyterLab header includes the title "LAT&PRT" and a "Last Checkpoint: 52 seconds ago" message. The interface has a menu bar with "File", "Edit", "View", "Run", "Kernel", "Settings", and "Help". Below the menu bar is a toolbar with icons for file operations and a "Code" button. The main area contains a code cell with the following Python code:

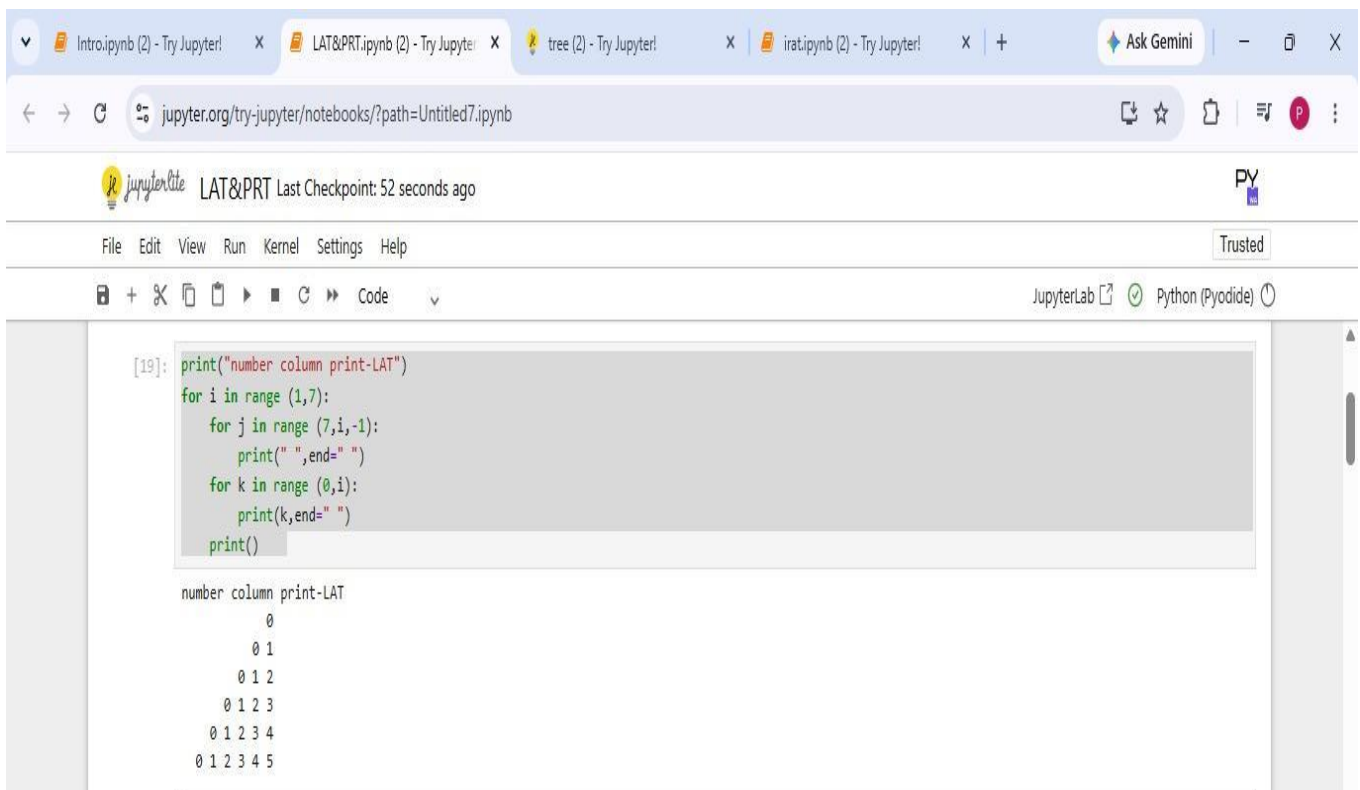
```
[11]: print("number row print-LAT")
      for i in range (1,7):
          for j in range (7,i,-1):
              print(" ", end=" ")
          for k in range (0,i):
              print(i,end=" ")
          print()
```

The output of the code is displayed below the code cell:

```
number row print-LAT
      1
     2 2
    3 3 3
   4 4 4 4
  5 5 5 5 5
 6 6 6 6 6 6
```

2.Left Angle Triangle (LAT) – Number- Column

Output



The screenshot shows a JupyterLab interface with a code cell and its output. The code cell contains the following Python code:

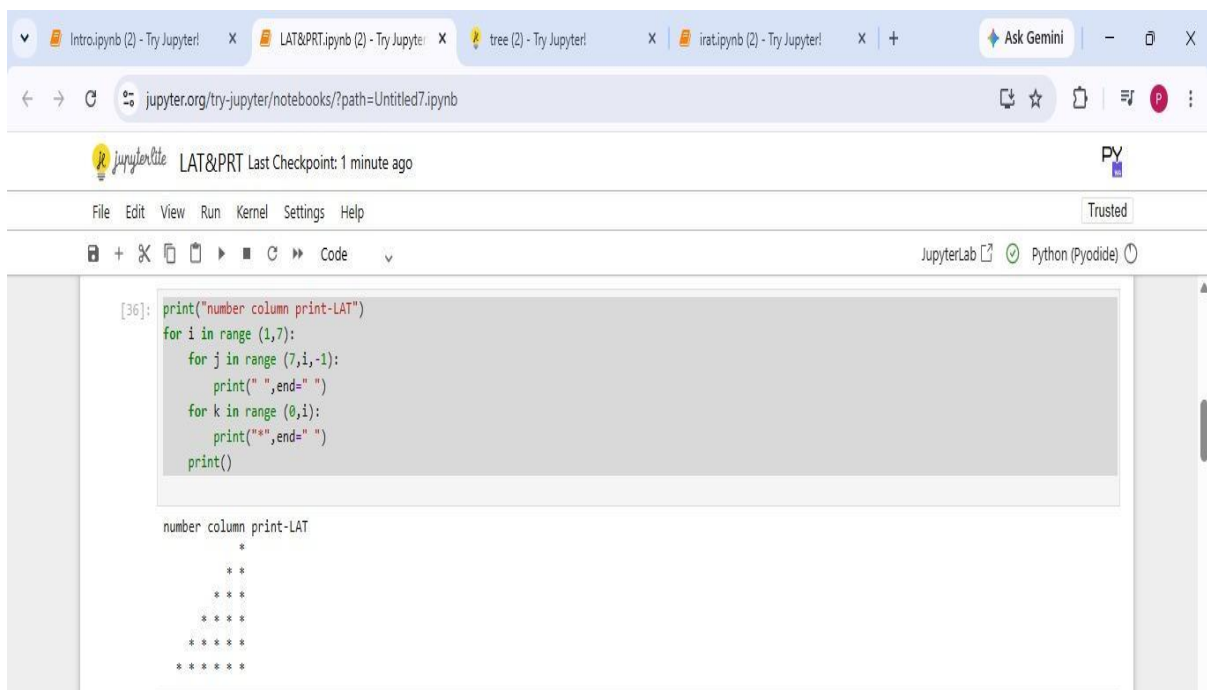
```
[19]: print("number column print-LAT")
for i in range(1,7):
    for j in range(7,i,-1):
        print(" ",end=" ")
    for k in range(0,i):
        print(k,end=" ")
    print()
```

The output of the code is a left-angled triangle of numbers:

```
number column print-LAT
      0
    0 1
  0 1 2
0 1 2 3
0 1 2 3 4
0 1 2 3 4 5
```

3.Left Angle Triangle (LAT)- STAR

Output



The screenshot shows a JupyterLab interface with a code cell and its output. The code cell contains the following Python code:

```
[36]: print("number column print-LAT")
for i in range(1,7):
    for j in range(7,i,-1):
        print(" ",end=" ")
    for k in range(0,i):
        print("*",end=" ")
    print()
```

The output of the code is a left-angled triangle of stars:

```
number column print-LAT
      *
    * *
  * * *
* * * *
* * * * *
* * * * *
```

4 .Left Angle Triangle (LAT)- upper alpha column

Output

The screenshot shows a JupyterLab window with a code cell containing the following Python code:

```
[30]: print("UPPER ALPHA column print-LAT")
for i in range(1,7):
    for j in range(7,i,-1):
        print(" ",end=" ")
    for k in range(0,i):
        print(chr(i+64),end=" ")
    print()
```

The output of the code is a Left Angle Triangle (LAT) with upper alpha column:

```
UPPER ALPHA column print-LAT
      A
     B B
    C C C
   D D D D
  E E E E E
 F F F F F F
```

5.Left Angle Triangle (LAT)- upper alpha Row

Output

The screenshot shows a JupyterLab window with a code cell containing the following Python code:

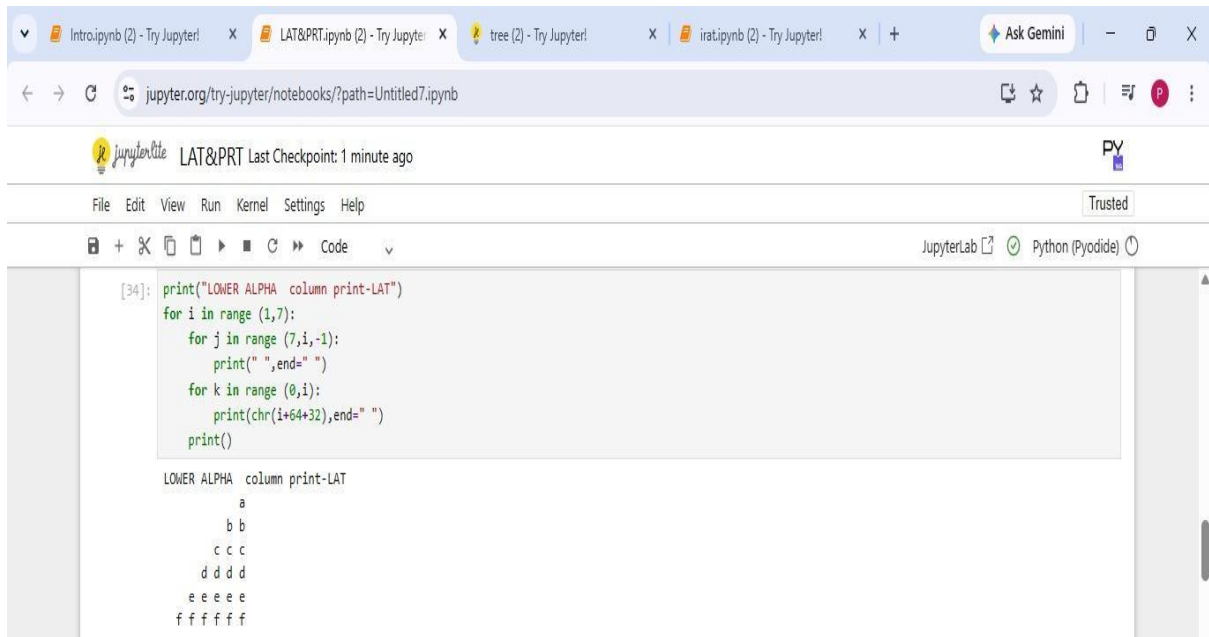
```
[33]: print("UPPER ALPHA row print-LAT")
for i in range(1,7):
    for j in range(7,i,-1):
        print(" ",end=" ")
    for k in range(0,i):
        print(chr(k+65),end=" ")
    print()
```

The output of the code is a Left Angle Triangle (LAT) with upper alpha row:

```
UPPER ALPHA row print-LAT
      A
     A B
    A B C
   A B C D
  A B C D E
 A B C D E F
```

6.Left Angle Triangle (LAT)- lower alpha Column

Output

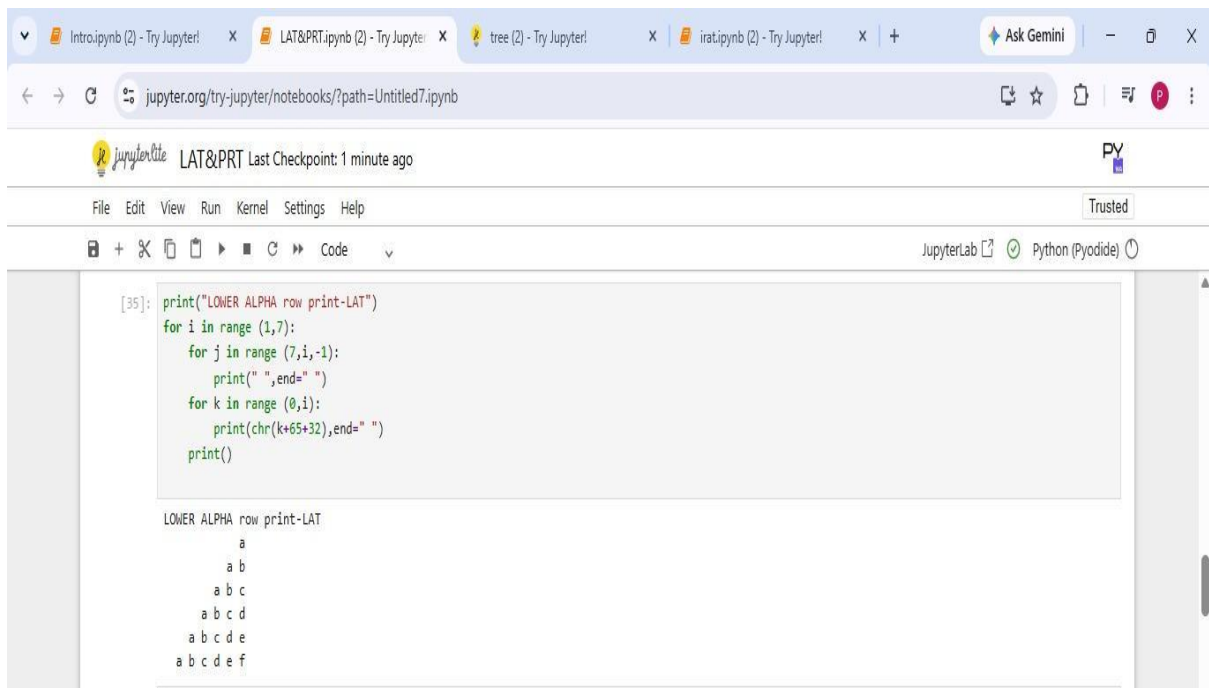


```
[34]: print("LOWER ALPHA column print-LAT")
      for i in range(1,7):
          for j in range(7,i,-1):
              print(" ",end=" ")
          for k in range(0,i):
              print(chr(i+64+32),end=" ")
          print()

LOWER ALPHA column print-LAT
      a
     b b
    c c c
   d d d d
  e e e e e
 f f f f f f
```

7.Left Angle Triangle (LAT)- lower alpha Row

Output

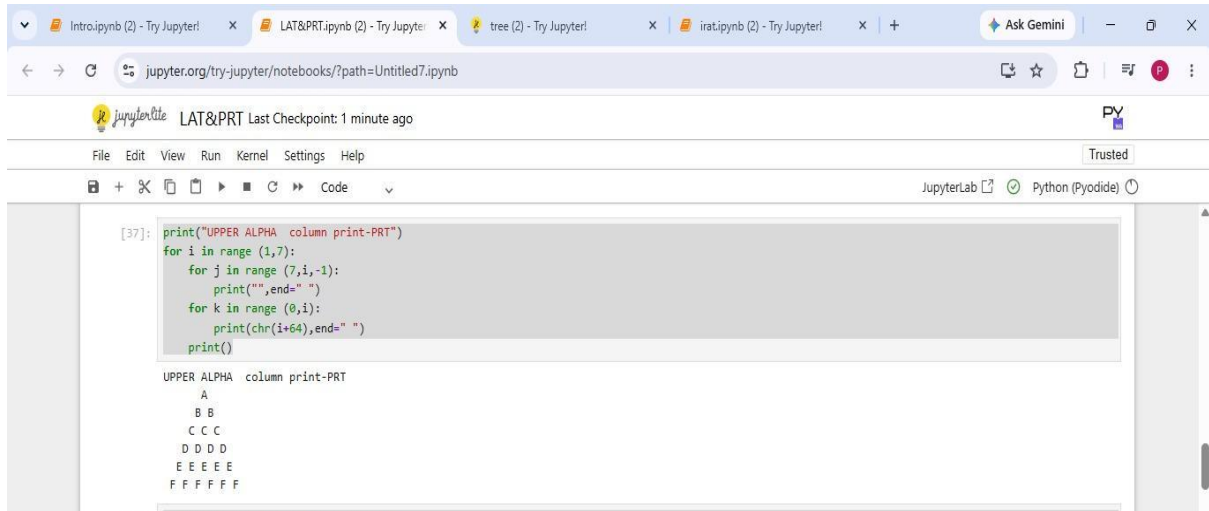


```
[35]: print("LOWER ALPHA row print-LAT")
      for i in range(1,7):
          for j in range(7,i,-1):
              print(" ",end=" ")
          for k in range(0,i):
              print(chr(k+65+32),end=" ")
          print()

LOWER ALPHA row print-LAT
      a
     a b
    a b c
   a b c d
  a b c d e
 a b c d e f
```

8.Pramid Triangle (PRT) – Upper Alpha column

Output



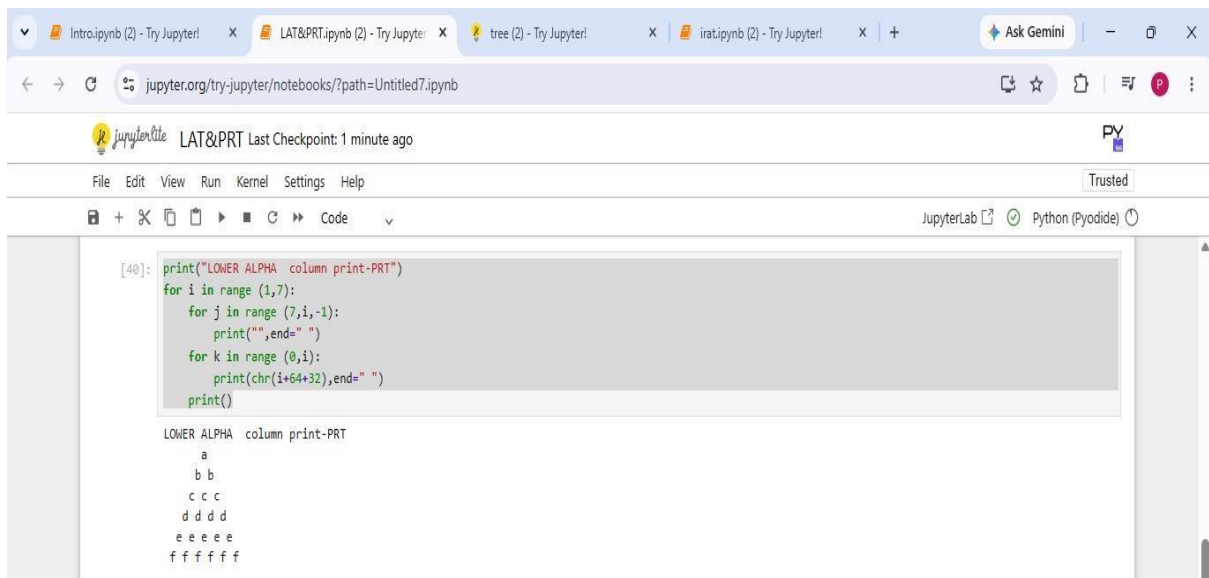
The screenshot shows a JupyterLab interface with a notebook titled "LAT&PRT". The code cell [37] contains a Python program that prints an upper alpha pyramid. The output is a pyramid of uppercase letters A through F, with each row containing a sequence of the same letter repeated i+1 times, where i is the row index (0 to 6).

```
[37]: print("UPPER ALPHA column print-PRT")
for i in range(1,7):
    for j in range(7,i,-1):
        print(" ",end=" ")
    for k in range(0,i):
        print(chr(i+64),end=" ")
    print()

UPPER ALPHA column print-PRT
A
B B
C C C
D D D D
E E E E E
F F F F F F
```

9.Pramid Triangle (PRT) – lower Alpha column

Output



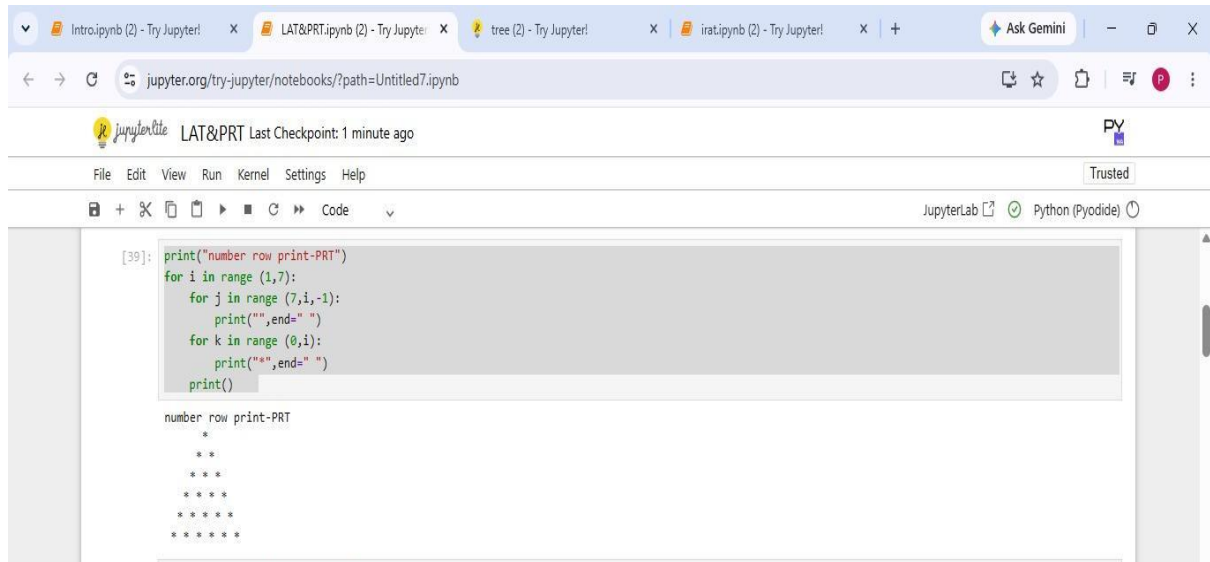
The screenshot shows a JupyterLab interface with a notebook titled "LAT&PRT". The code cell [40] contains a Python program that prints a lower alpha pyramid. The output is a pyramid of lowercase letters a through f, with each row containing a sequence of the same letter repeated i+1 times, where i is the row index (0 to 6).

```
[40]: print("LOWER ALPHA column print-PRT")
for i in range(1,7):
    for j in range(7,i,-1):
        print(" ",end=" ")
    for k in range(0,i):
        print(chr(i+96),end=" ")
    print()

LOWER ALPHA column print-PRT
a
b b
c c c
d d d d
e e e e e
f f f f f f
```

10.Pramid Triangle (PRT) – STAR

Output

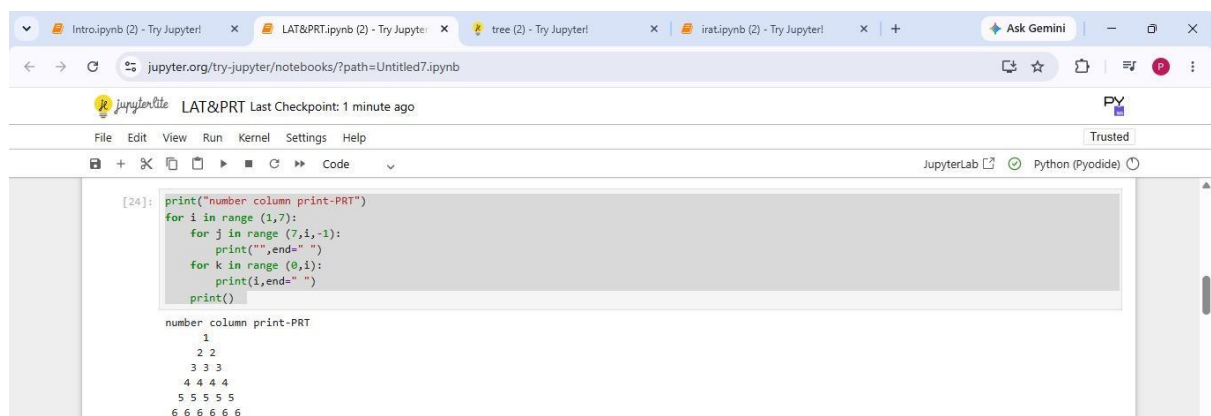


```
[39]: print("number row print-PRT")
for i in range(1,7):
    for j in range(7,i,-1):
        print("",end=" ")
    for k in range(0,i):
        print("*",end=" ")
    print()

number row print-PRT
*
* *
* * *
* * * *
* * * * *
* * * * *
* * * * *
```

11. Pramid Triangle (PRT) – Number column

Output

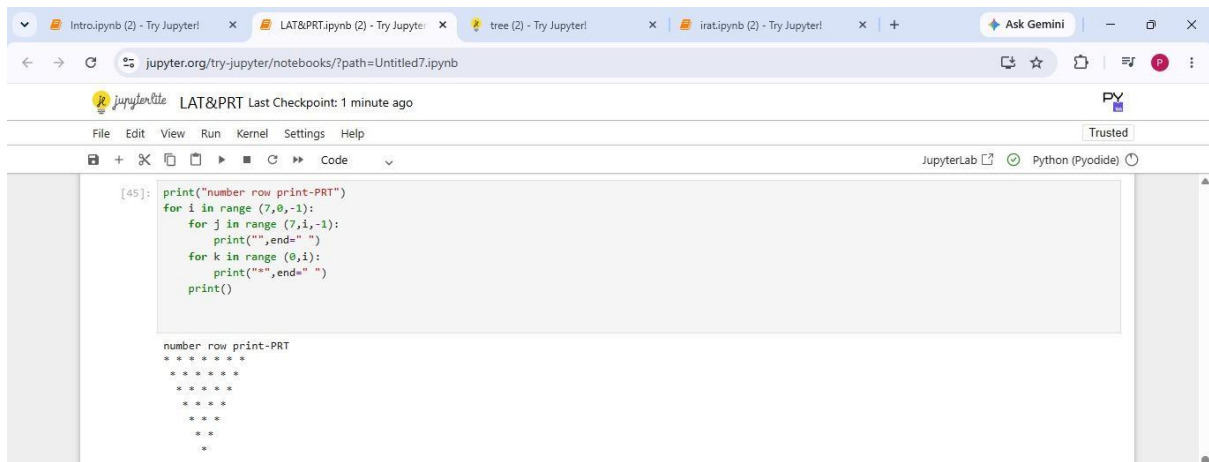


```
[24]: print("number column print-PRT")
for i in range(1,7):
    for j in range(7,i,-1):
        print("",end=" ")
    for k in range(0,i):
        print(i,end=" ")
    print()

number column print-PRT
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
6 6 6 6 6 6
```

12.Inverse Pramid Triangle (IPRT) – Number- Row

Output

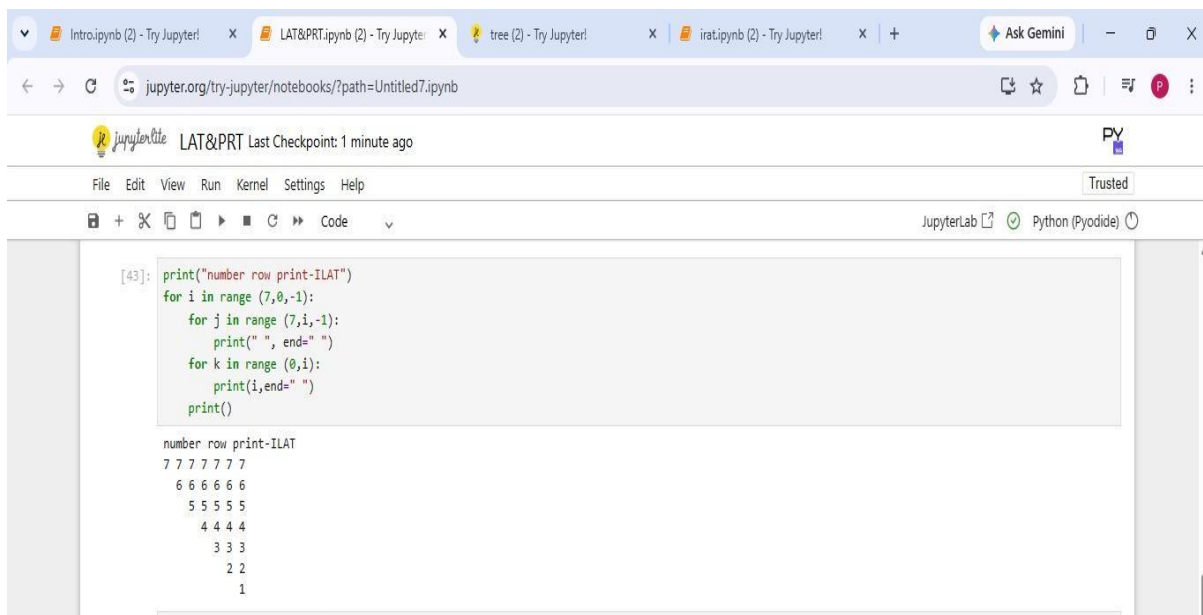


```
[45]: print("number row print-PRT")
      for i in range (7,0,-1):
          for j in range (7,i,-1):
              print("",end=" ")
          for k in range (0,i):
              print("=",end=" ")
          print()

number row print-PRT
*****
*****
*****
*****
*****
*****
*****
```

13.Inverse Left Angle triangle (ILAT) – Number Row

Output

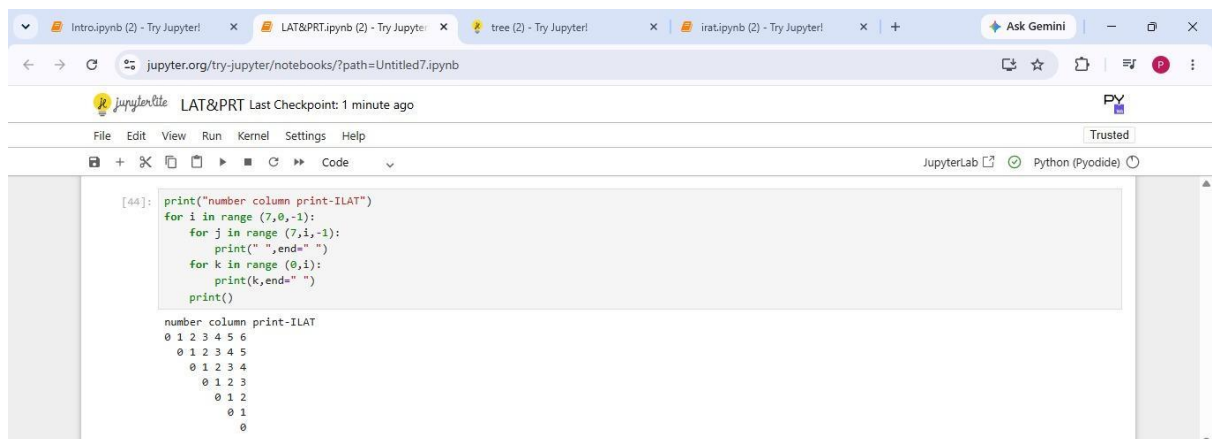


```
[43]: print("number row print-ILAT")
      for i in range (7,0,-1):
          for j in range (7,i,-1):
              print(" ", end=" ")
          for k in range (0,i):
              print(i,end=" ")
          print()

number row print-ILAT
7 7 7 7 7 7
6 6 6 6 6
5 5 5 5
4 4 4 4
3 3 3
2 2
1
```

14. Inverse Left Angle Triangle (ILAT) – Number Column

Output

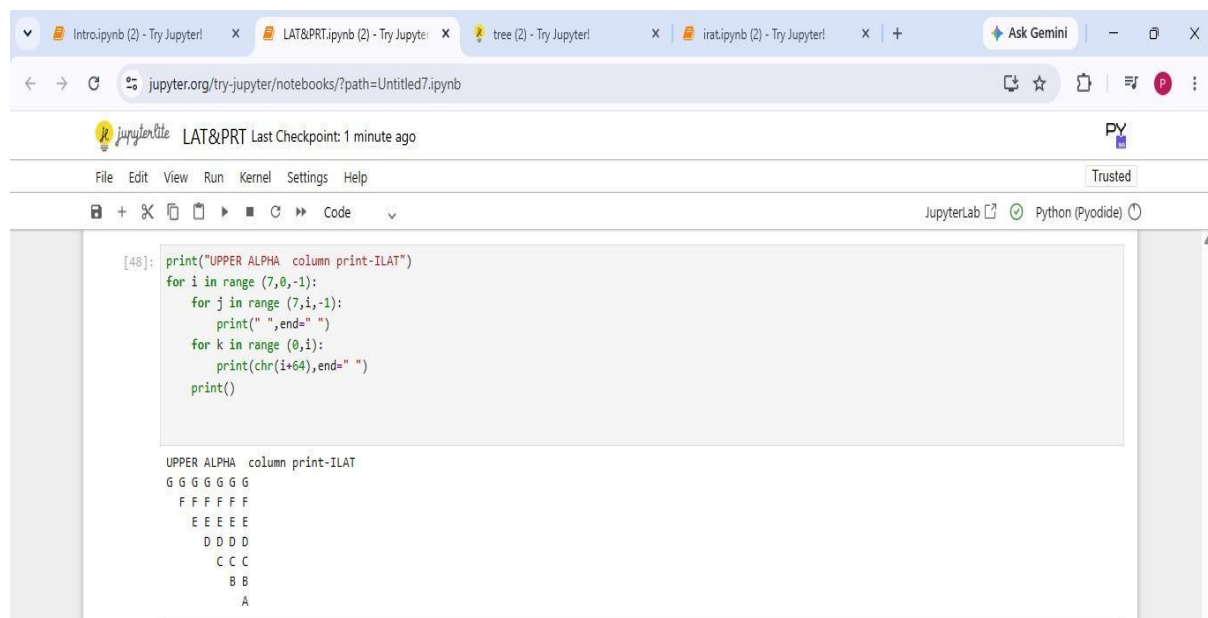


```
[44]: print("number column print-ILAT")
      for i in range(7,0,-1):
          for j in range(7,i,-1):
              print(" ",end=" ")
          for k in range(0,i):
              print(k,end=" ")
          print()

number column print-ILAT
0 1 2 3 4 5 6
0 1 2 3 4 5
0 1 2 3 4
0 1 2 3
0 1 2
0 1
0
0
```

15. Inverse Left Angle Triangle (ILAT) – Upper Alpha column

Output

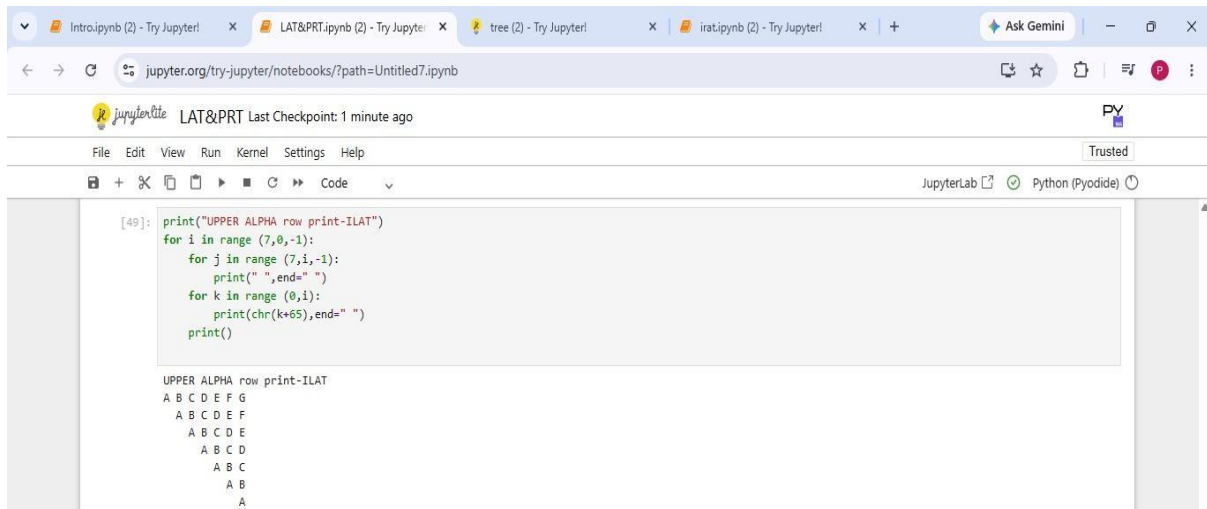


```
[48]: print("UPPER ALPHA column print-ILAT")
      for i in range(7,0,-1):
          for j in range(7,i,-1):
              print(" ",end=" ")
          for k in range(0,i):
              print(chr(i+64),end=" ")
          print()

UPPER ALPHA column print-ILAT
G G G G G G
F F F F F F
E E E E E
D D D D
C C C
B B
A
```

16. Inverse Left Angle Triangle (ILAT) – Upper Alpha Row

Output



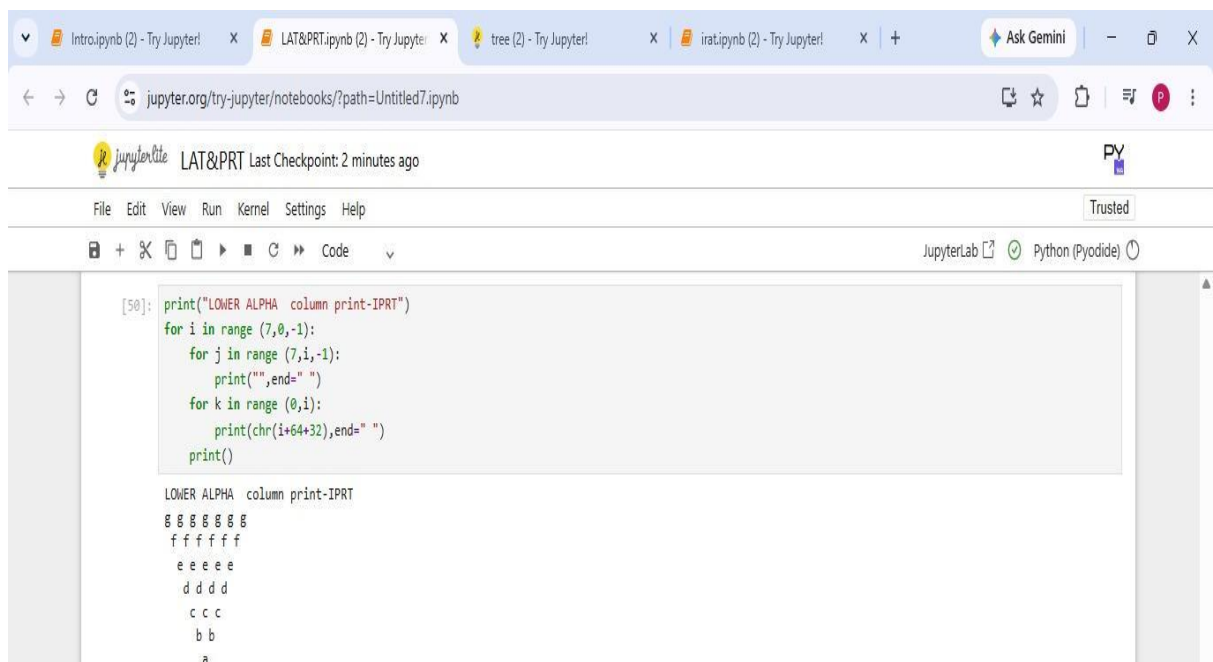
The screenshot shows a JupyterLab interface with a code editor and an output area. The code editor contains a Python program that prints an upper alpha row pyramid. The output area shows the result of the program, which is an upper alpha row pyramid with 7 rows and 7 columns.

```
[49]: print("UPPER ALPHA row print-ILAT")
      for i in range(7,0,-1):
          for j in range(7,i,-1):
              print(" ",end=" ")
          for k in range(0,i):
              print(chr(k+65),end=" ")
          print()
```

UPPER ALPHA row print-ILAT
A B C D E F G
A B C D E F
A B C D E
A B C D
A B C
A B
A

17.Inverse Pramid Triangle (IPRT) – lower alpha column

Output



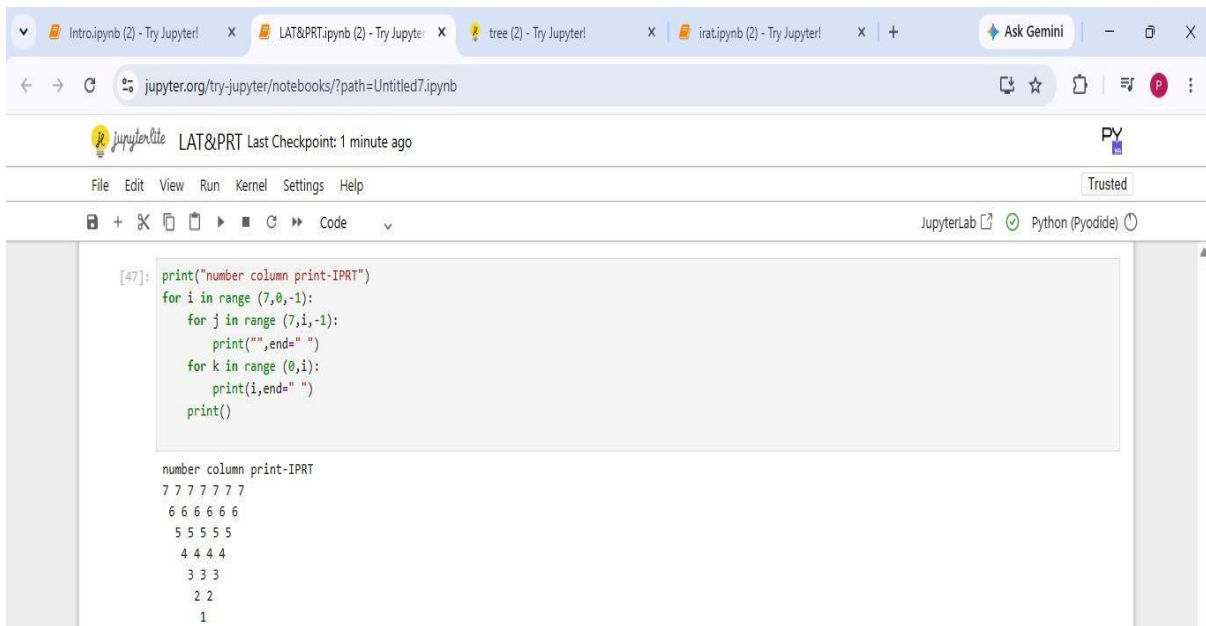
The screenshot shows a JupyterLab interface with a code editor and an output area. The code editor contains a Python program that prints a lower alpha column pyramid. The output area shows the result of the program, which is a lower alpha column pyramid with 7 rows and 7 columns.

```
[50]: print("LOWER ALPHA column print-IPRT")
      for i in range(7,0,-1):
          for j in range(7,i,-1):
              print("",end=" ")
          for k in range(0,i):
              print(chr(i+64+32),end=" ")
          print()
```

LOWER ALPHA column print-IPRT
g g g g g g g
f f f f f
e e e e
d d d
c c c
b b
a

18.Inverse Pramid Triangle (IPRT) – number column

Output

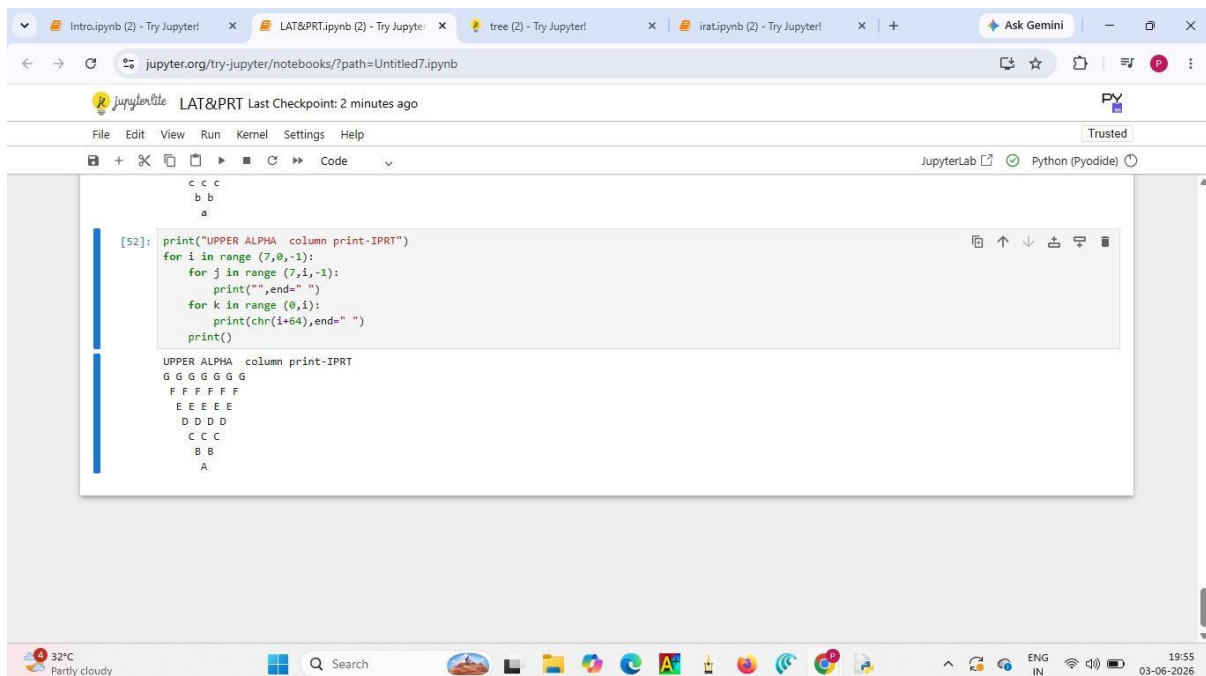


```
[47]: print("number column print-IPRT")
for i in range (7,0,-1):
    for j in range (7,i,-1):
        print("",end=" ")
    for k in range (0,i):
        print(i,end=" ")
    print()
```

number column print-IPRT
7 7 7 7 7 7
6 6 6 6 6
5 5 5 5
4 4 4 4
3 3 3
2 2
1

19.Inverse Pramid Triangle – Upper Alpha column

Output

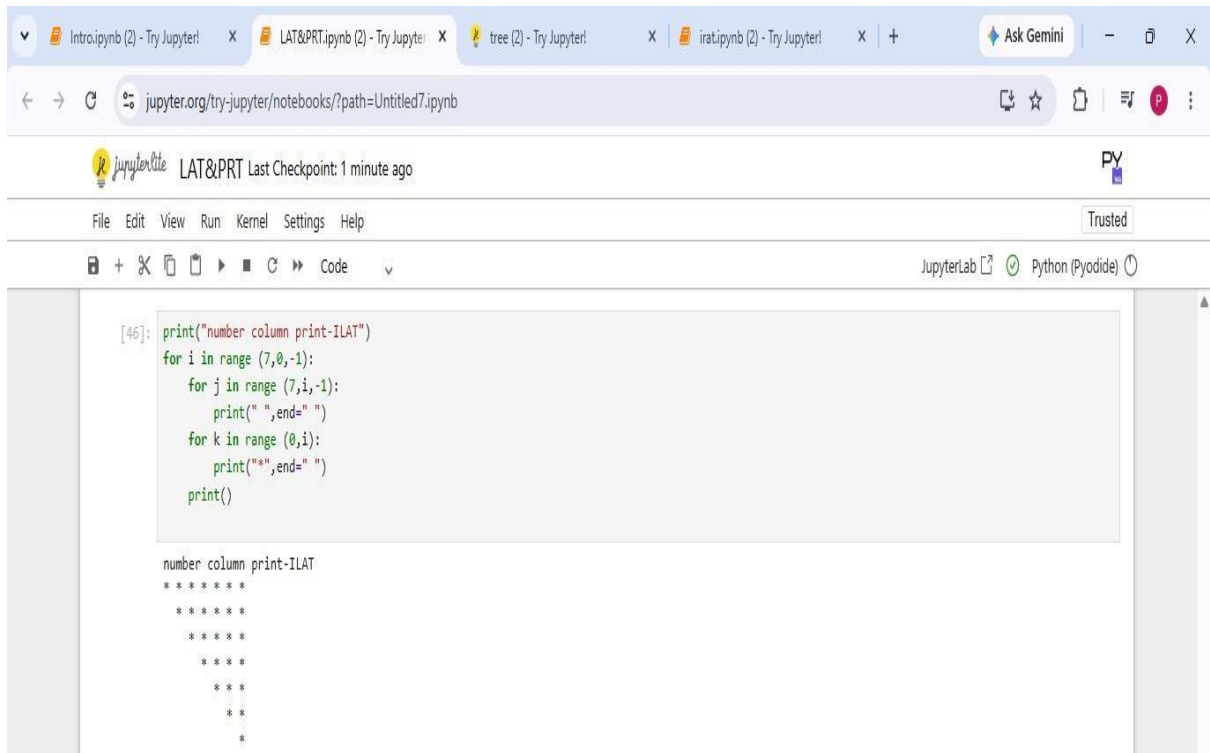


```
[52]: print("UPPER ALPHA column print-IPRT")
for i in range (7,0,-1):
    for j in range (7,i,-1):
        print("",end=" ")
    for k in range (0,i):
        print(chr(i+64),end=" ")
    print()
```

UPPER ALPHA column print-IPRT
G G G G G G
F F F F F F
E E E E E
D D D D
C C C C
B B
A

20. Inverse Left Angle Triangle – Number Column

Output



The screenshot shows a JupyterLab interface with a browser window at the top. The browser has several tabs: 'Intro.ipynb (2) - Try Jupyter!', 'LAT&PRT.ipynb (2) - Try Jupyter!', 'tree (2) - Try Jupyter!', and 'irat.ipynb (2) - Try Jupyter!'. The active tab is 'LAT&PRT.ipynb (2) - Try Jupyter!'. The address bar shows 'jupyter.org/try-jupyter/notebooks/?path=Untitled7.ipynb'. Below the browser, the JupyterLab interface is visible. The top bar shows 'JupyterLab' and 'Python (Pyodide)'. The main area contains a code cell with the following Python code:

```
[46]: print("number column print-ILAT")
      for i in range (7,0,-1):
          for j in range (7,i,-1):
              print(" ",end=" ")
          for k in range (0,i):
              print("*",end=" ")
          print()
```

The output of the code is an inverse left angle triangle pattern of asterisks:

```
number column print-ILAT
* * * * *
 * * * * *
  * * * *
   * * *
    * *
     *
```